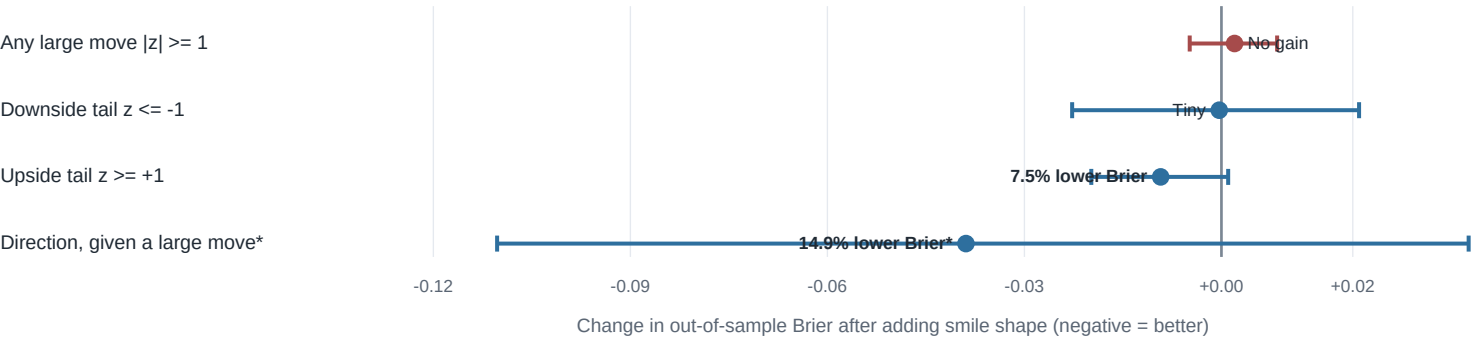


Options as anticipative factors: where does the smile's information live?

Exploratory Phase II note - NIFTY weekly options, one non-overlapping observation per expiry

Question. After conditioning on the ATM option-implied movement scale, does cross-strike smile geometry add information about the size of the future move, or about which side of the distribution is tilted?

A. Incremental predictive content of smile geometry



B. Width robustness: direction is more stable than magnitude



Compact out-of-sample readout

Task	N/events	Brier level	Brier +shape	Delta	AUC level -> shape	Widths
Any one-scale breach	73 / 23	0.2268	0.2288	+0.0020	0.501 -> 0.493	0/8
Downside breach	73 / 13	0.1564	0.1561	-0.0003	0.574 -> 0.596	4/8
Upside breach	73 / 10	0.1227	0.1134	-0.0092	0.605 -> 0.635	8/8
Direction given breach*	23 / 13	0.2632	0.2241	-0.0391	0.631 -> 0.685	8/8

Readout. Smile shape does not improve two-sided one-scale breach probability (0/8 widths). The strongest OOS signal is directional: upside-breach Brier falls 0.1227 -> 0.1134, and all 8/8 widths improve it. Conditional direction also improves at 8/8 widths, but that N=23 result conditions on a future breach and is diagnostic.

Interpretation. The pattern is consistent with ATM level summarizing forward risk magnitude while cross-strike state prices may encode distributional tilt. This is hypothesis-generating, not conclusive; next: θ vs $\theta + \{\rho, \psi, RR, BF\}$.